

m-Zephyr: A Digital In-Memory Ising Chip with 240 Spins Featuring Enhanced Connectivity Based on a Modified 3D Zephyr Topology

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Abstract

We present a digital in-memory Ising computing chip with a modified 3D Zephyr topology implemented using a 65nm CMOS technology. This design features cross-dimensional connectivity with 24 interactions per spin for the efficient mapping of complex combinatorial optimization problems to the Ising hardware. The proposed Ising chip with enhanced spin connectivity outperforms prior works in terms of reconfigurability, allowing the Ising chip to solve more difficult problems with complicated topologies and higher dimensions, but minimizing hardware mapping overhead. $24\times$ 7-bit precision coefficients are stored in memory and used to compute spin interactions, minimizing energy consumption and computing latency while ensuring accuracy with fully digital and high precision computing. Measured results show $>100\times$ speedup than the baseline algorithm while finding better optimal solutions with lower Mean Ising Hamiltonian values.

Introduction

Combinatorial optimization problems are encountered in broad industries such as logistics, telecommunications, manufacturing, and finance. These problems are categorized as non-deterministic polynomial-time hard (NP-hard) due to exponentially growing complexity as the problem size increases. Hardware accelerators based on a mathematical model of ferromagnetism, Ising model, have recently gained notable interest thanks to the capability of solving NP-hard problems much faster than classical computers using the natural fast convergence of magnetic spin states through massive parallel spin interactions. Among various Ising hardware implementations, Ising chips fabricated using CMOS technology have quickly become popular due to the technology maturity and better scalability than other technologies.

Prior works [1-9] reported digital and mixed-signal Ising chips, but these early Ising chips have their own drawbacks. Most prior works implemented simpler spin-processing element arrays with simpler locally connected hardware topologies, such as lattice [1-2] and King's graph [3-5]. Due to limited spin connectivity, they showed limited capabilities of mapping more complicated real-world problems to the Ising chip. A fully connected Ising chip [6] addressed the reconfigurability issue by implementing all-to-all spin connectivity, but they are inefficient due to lower utilization of their available hardware connectivity, as real-world problems are often coarser than full connections. Other works [7-8] focused more on improving solution time than hardware programmability. They achieved a notable speed-up but were eventually limited to impractically small problems due to simple hardware topologies. Enhanced chimera topology [9] has recently been introduced with improved connectivity (11 connections per spin), but their ternary interaction precision is too low, and achieving higher precision in an analog resistive coupling is a challenging task.

Proposed 3D Reconfigurable Ising Machine

In this work, we propose a digital in-memory computing Ising chip for enhanced connectivity and reconfigurability based on the modified 3D Zephyr topology we implemented to solve complex Ising problems, as shown in Fig. 1. There are 24 reconfigurable connections per spin to map and solve various NP-hard problems. The proposed chip not only improves reconfigurability, but it also achieves state-of-the-art Ising chip performance with no accuracy degradation by implementing 7-bit in-memory spin interactions that are fully digital, achieving low energy consumption ($<350\text{nJ}$) and fast solution time ($<180\text{ns}$). Key features are listed in Fig. 2.

Fig. 3 shows the structure of the proposed 3D m-Zephyr model, in which spins are connected across stacked layers, enabling more flexible inter-layer interactions than two-dimensional (2D) Ising topologies. An example is a $4\times 4\times 4$ configuration including 64 spins, grouped into four layers (XY planes) of 16 spins each. The

left panel labeled 'A' shows partial interactions in a YZ plane, in which four spins stacked vertically form a group with no coupling between them while fully connected to spins in adjacent groups. Similarly, the right panel labeled 'B' shows the same interactions but in an XZ plane. Besides the two inter-plane spin connectivity, which already forms 16 connections per spin, additional coupling (8 connections per spin) forms between groups that are located in diagonal directions, as shown in Fig. 3 (bottom-left). Fig. 3 (right) shows a simplified view with all spins projected onto the XY plane, expanding every vertically stacked spin group in the same 2D topology, retaining the original 3D connectivity. Embedding the expanded m-Zephyr topology onto a chip is shown in Fig. 4.

Fig. 5 shows the detailed block diagram of the proposed 65 nm reconfigurable digital Ising chip architecture, which implements a $6\times 10\times 4$ 3D m-Zephyr topology on a two-dimensional chip that comprises 240 spins with 41kb SRAM for interaction coefficients stored and computed in memory. Each spin cell contains top and bottom interacting buses for massively parallel interactions with spins based on m-Zephyr topology. Fig. 6 shows the core circuit, where 6 spin interactor units compute the partial gradient of Ising Hamiltonian (H) for group interactions from all six directions based on the m-Zephyr topology. Each integrator unit comprises four sub-blocks responsible for a full connection with four spins from a group that a spin interacts with. As shown in Fig. 6 (right), a sub-block consists of 7-bit SRAM, XNOR gates, and full adders for a bit-serial multiply-and-add operation. The computed partial gradient Hamiltonian values propagate to a spin logic, where the spin state is updated based on the sign of the accumulated partial gradient Ising Hamiltonian values. If the partial sum equals zero, the spin state is inverted, adding an extra randomness that helps find better sub-optimal solutions.

Fig. 7 illustrates how we reconfigure the proposed chip to map various Ising model topologies using a multi-layer stacked lattice graph as an example. The multi-layer lattice topology is expanded, repositioned, and finally mapped onto the chip. Besides the lattice topology, more complicated topologies can be mapped onto the proposed m-Zephyr topology with minimal hardware overhead by following the same strategy to map complex spin connections onto different layers of 3D m-Zephyr topology.

Fig. 8 (top) shows the chip test setup and overall workflow. Fig. 8 (bottom) illustrates an easy Max-Cut problem with a predefined ground state. The spin states quickly converge to a spin-up (white pixel) or a spin-down (black pixel) to show a clear 'ISING' image when the chip converges to a ground state with the minimal Ising Hamiltonian value. Transient spin state maps and corresponding Ising Hamiltonian values are shown.

Fig. 9 shows more extensive measurement results when solving hard Max-Cut problems, where random interaction coefficients (ranging from -63 to +63 using full 7-bit coefficients) are mapped to all $24\times$ interaction coefficients for 240 spins. A total of 1,000 hard Max-Cut problems are solved. Measured results show $\sim 10^6\times$ faster solution time for the proposed chip than the sequential Ising solving using CPU while operating at 166.66MHz. Overall, a fast solution time of $<180\text{ns}$ and a low energy consumption of $<350\text{nJ}$ has been achieved consistently over supply voltage ranges (0.8 to 1.2V) while lowering the Mean Ising Hamiltonian value than the baseline Metropolis algorithm by 4.13%.

Fig. 10 summarizes the measured results, including a plot with measured energy consumptions (217 to 350nJ) when the supply voltage is swept from 0.8 to 1.2V. A chip summary table is shown in Fig. 10 (top left). A die micrograph of the fabricated 65nm test chip is shown in Fig. 10 (right), where a core spin array (240 spins with $24\times$ 7-bit interactions per each) occupies 1.1565mm^2 . Table I compares the proposed work with state-of-the-art Ising chips.

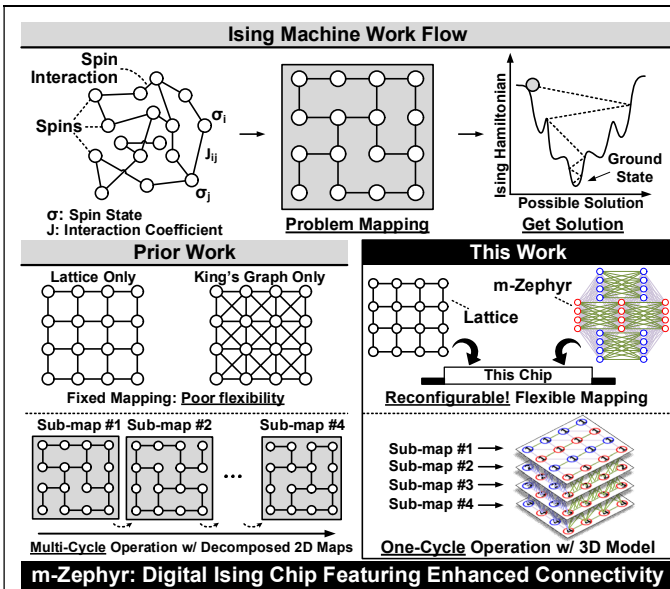


Fig. 1. Work flow of an Ising Machine (top), and comparison between prior work (bottom left) and this work (bottom right).

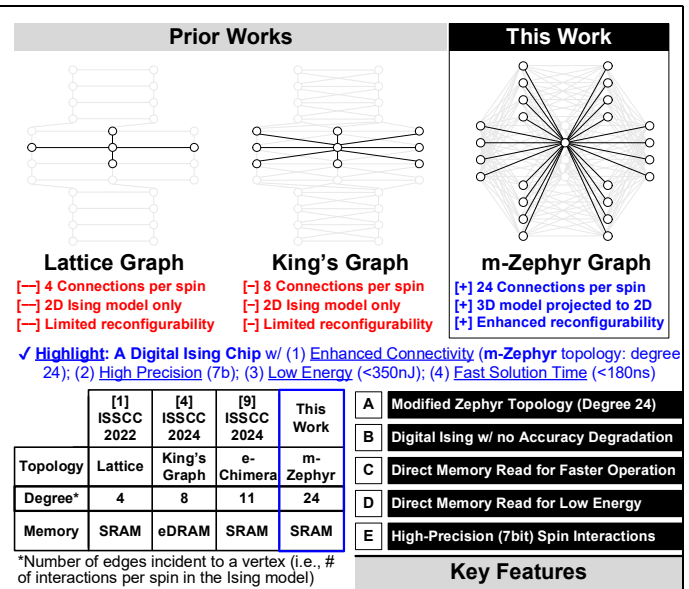


Fig. 2. Proposed m-Zephyr graph, comparison to prior graphs (top and bottom left), and key features (bottom right).

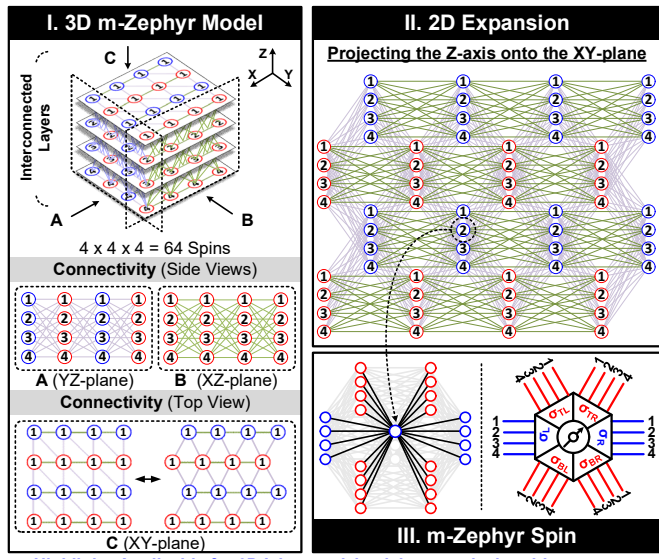


Fig. 3. Proposed 3D m-Zephyr model (left) and its expansion to a 2D m-Zephyr Ising chip spin array structure (right).

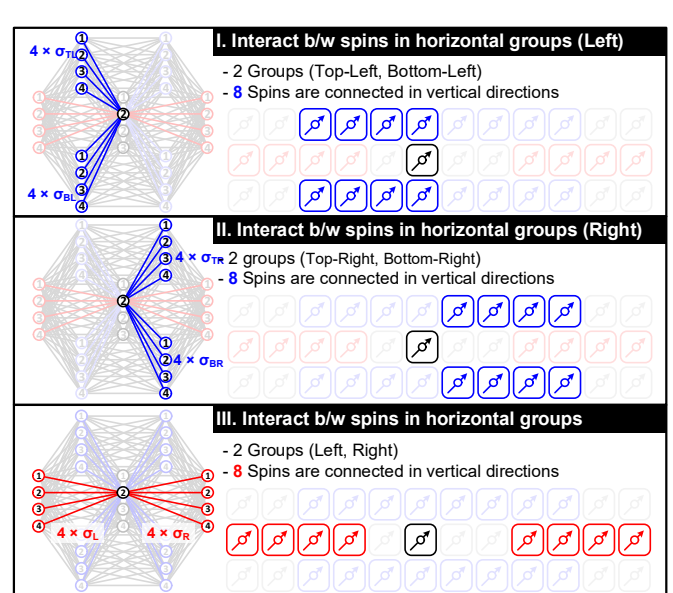


Fig. 4. Embedding the m-Zephyr model to the proposed Ising chip.

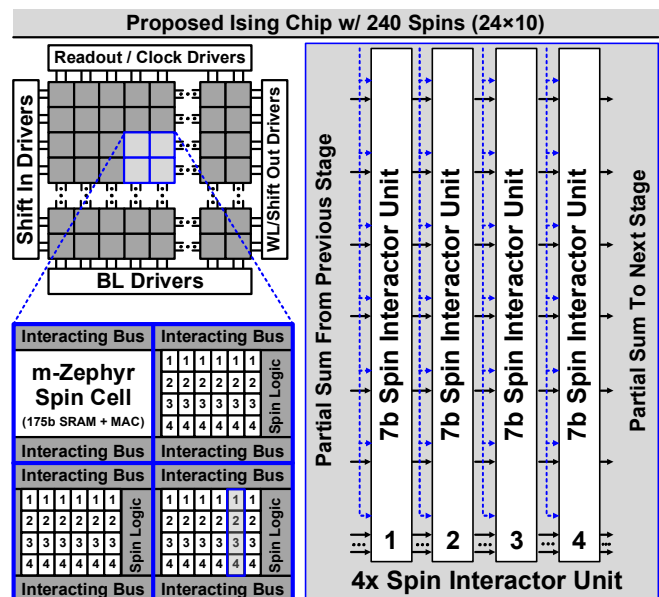


Fig. 5. Overall architecture of the m-Zephyr Ising chip.

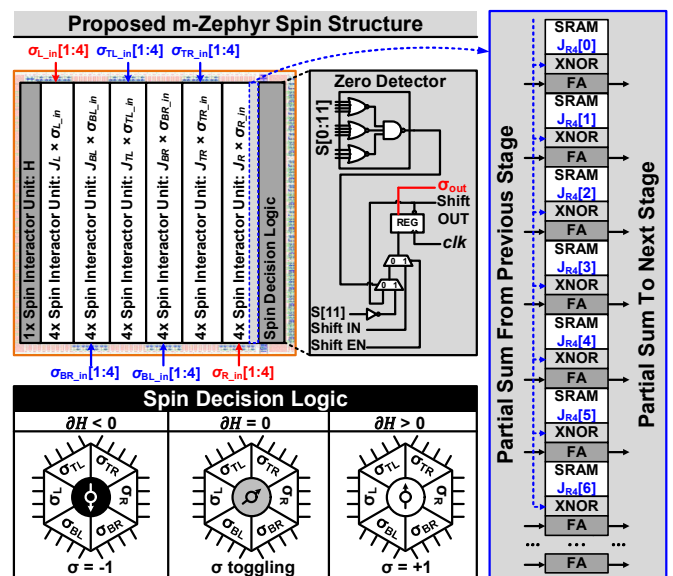


Fig. 6. Schematic diagram of the m-Zephyr spin circuit and its spin state decision logic.

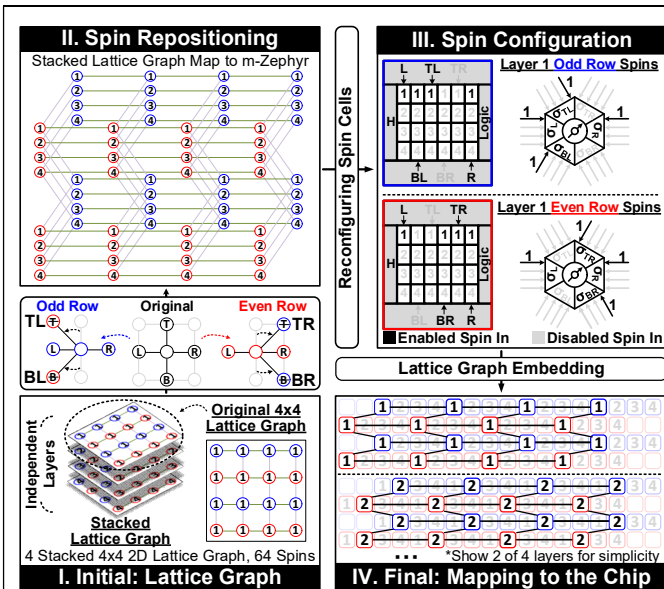


Fig. 7. A multi-layer stacked lattice topology mapped to the proposed m-Zephyr Ising chip.

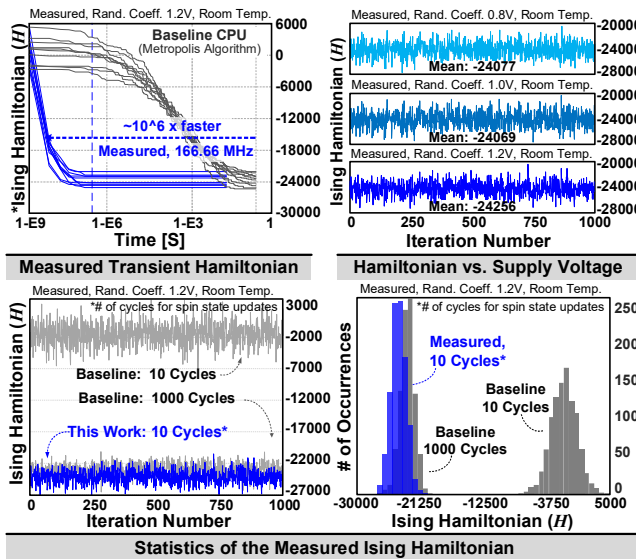


Fig. 9. Measured hard Max-Cut problems. Transient results (top-left), Hamiltonian vs. supply voltage (top-right), and statistics (bottom).

Table I. Comparison with state-of-the-art ASIC Ising chips.

	VLSI 2021 [5]	JSSC 2022 [3]	ISSCC 2024 [9]	This Work
Technology (Circuit Type)	65nm CMOS (Mixed-Signal)	65nm CMOS (Mixed-Signal)	65nm CMOS (Mixed-Signal)	65nm CMOS (Digital)
Operating Temperature	Room Temp. (300K)	Room Temp. (300K)	Room Temp. (300K)	Room Temp. (300K)
Spin Memory	Register	eDRAM Cell	SRAM Cell	Register
Spin State Representation	Analog (Latch Voltage)	Digital (Binary State)	Analog (Latch Voltage)	Digital (Binary State)
Spin Interaction	Analog Accumulate	Mixed-Signal MAC	Latch Coupling Via Switches	Digital MAC
Graph Topology	King's Graph	King's Graph	e-Chimera	m-Zephyr ^A
3D Ising Model Applicable	No	No	No	Yes
# of Interactions Per Spin	8	8	11	24
SRAM R/W During Ising	Directly Read Only (No Write)	Required (via SRAM I/O)	Required (via SRAM I/O)	Directly Read Only (No Write)
Coefficient Bitwidth	4bit	1-4bit	2bit	7bit
Core Voltage	0.75-1.05V	0.9-1.2V	0.8-1.4V	0.8-1.2V
Solution Time	100-3000ns	51200 ns (12800 Cycles)	<100 ns	<180ns

^AModified Zephyr topology with enhanced connectivity (degree of 24)

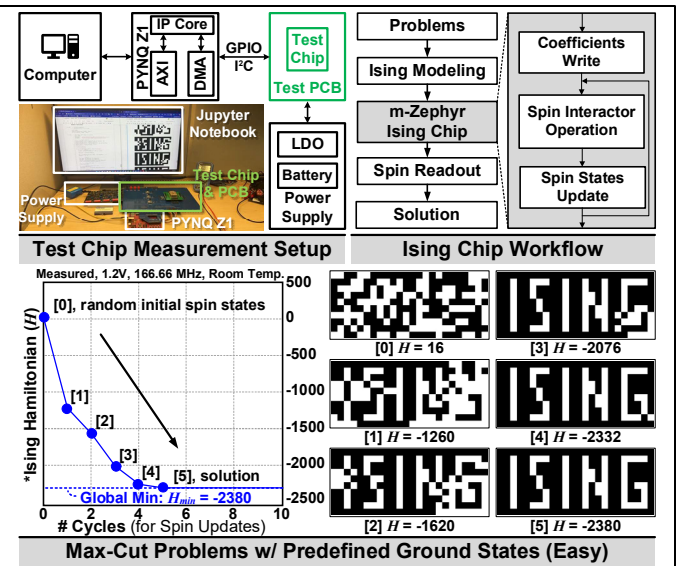


Fig. 8. Measurement setup (top-left), Ising chip workflow (top-right), and easy Max-Cut problems mapping to the proposed chip, where the 'ISING' pattern appears as the ground state (bottom).

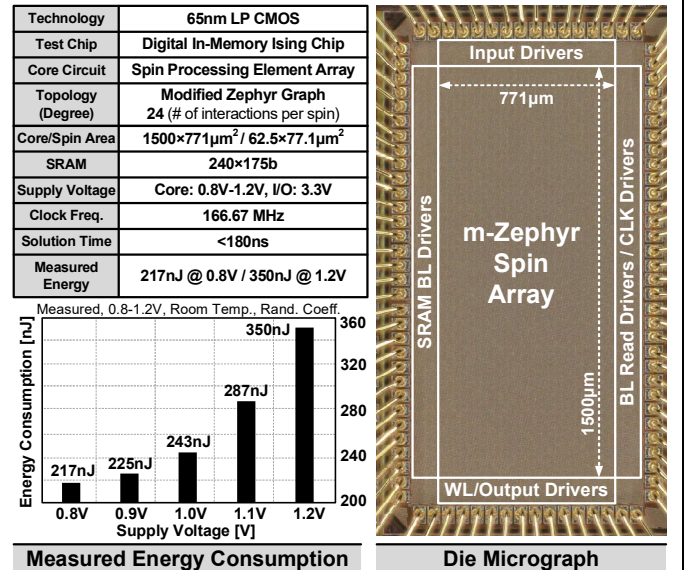


Fig. 10. Test chip summary (top-left), measured energy consumption (bottom-left), and test chip die micrograph (right).

References

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